

PYTHON LAB EXERCISE 7

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1. Write a python program that creates a pandas dataframe from scratch using a dictionary display the dataframe and then add a new column with calculated values:

```
import pandas as pd
data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'David'],
    'Age': [25, 30, 35, 40],
    'Salary': [50000, 60000, 70000, 80000]
}

df = pd.DataFrame(data)
print("--- Original DataFrame ---")
print(df)
df['Bonus'] = df['Salary'] * 0.10
df['Total_Comp'] = df['Salary'] + df['Bonus']
print("\n--- Updated DataFrame with Bonus ---")
print(df)
```

```
--- Original DataFrame ---
   Name  Age  Salary
0  Alice   25   50000
1   Bob    30   60000
2 Charlie   35   70000
3  David   40   80000

--- Updated DataFrame with Bonus ---
   Name  Age  Salary  Bonus  Total_Comp
0  Alice   25   50000  5000.0   55000.0
1   Bob    30   60000  6000.0   66000.0
2 Charlie   35   70000  7000.0   77000.0
3  David   40   80000  8000.0   88000.0
```

2. Read a CSV file as dataframe and display first 6 lines and basic info using info():

```
import pandas as pd

df = pd.read_csv("C:\\Users\\RVUW289\\Downloads\\coco - Sheet1.csv")

print(df.head(6))

df.info()
```

	Employee ID	Full Name	Department	Hire Date	Salary	Status
0	101	Alice Johnson	HR	2021-03-15	65,000	Active
1	102	Bob Smith	Engineering	2020-01-10	95,000	Active
2	103	Charlie Davis	Engineering	2022-06-01	88,000	On Leave
3	104	Diana Prince	Marketing	2019-11-20	72,000	Active
4	105	Ethan Hunt	Sales	2023-02-14	58,000	Active
5	106	Fiona Glenanne	HR	2021-07-22	67,000	Resigned

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Employee ID     6 non-null     int64
1   Full Name       6 non-null     object
2   Department      6 non-null     object
3   Hire Date       6 non-null     object
4   Salary          6 non-null     object
5   Status          6 non-null     object
dtypes: int64(1), object(5)
memory usage: 420.0+ bytes
```